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Joby Aviation and defense contractor L3Harris to test autonomous hybrid aircraft

Joby



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Joby Aviation is often cast as a developer of commercial electric air taxis, but the publicly traded company has also pursued a separate track to market through a long-standing relationship with the U.S. Department of Defense. Now, its years of research and development with the Department of Defense may be paying off.

The company on Thursday said it had signed an agreement with defense contractor L3Harris Technologies to “explore opportunities” to develop a new aircraft class — specifically, a gas-turbine hybrid vertical take-off and landing aircraft that can fly autonomously — for defense applications.

IMAGE CREDITS: JOBY AVIATION

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The gas-turbine hybrid VTOL will be based on Joby’s current S4 aircraft platform. The company has focused on developing the S4 with an all-electric powertrain, but it last year demonstrated under government contract a [hydrogen-electric hybrid version](#) that flew 521 miles — more than two times farther than its battery electric prototype.

The agreement with L3Harris is exploratory, but the hope is that upcoming flight tests and operational demonstrations will lead to a military contract with the DoD. Joby said flight testing will start this fall,

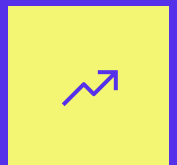
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and operational demonstrations are expected in 2026.

Joby has spent years developing and seeking Type 1 certification from the Federal Aviation Administration for its all-electric vertical take-off and landing aircraft, which will be used to shuttle passengers short distances in urban areas.

It's also worked with the DoD at the development stage for nearly a decade. Through that relationship, Joby homed in on what it would take to get to the "finish line" with the DoD, Joby executive chairman Paul Sciarra told TechCrunch.

"One was that range was especially important," he said. "We knew we had to make that shift."

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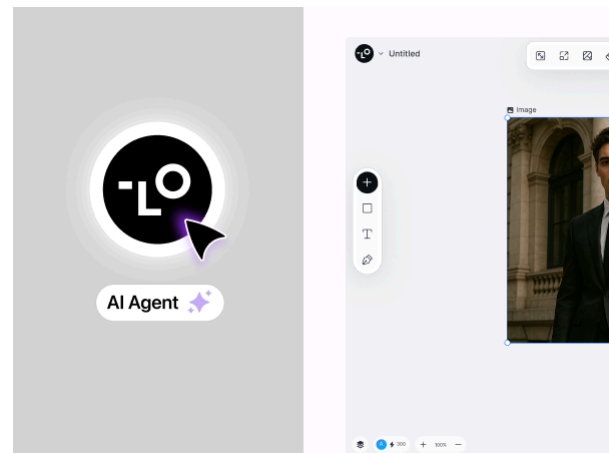
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Sciarra added it was also important to demonstrate the so-called "missionization of its platform" — how its aircraft could be combined with the right sensors, autonomy, communications and payload to provide a clear use case for the DoD.

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“We have two choices: either build a bunch of this stuff ourselves, or find an incredible partner that already has a deep understanding of those use cases and a ton of proven technology,” he said.

That’s where L3Harris comes in.

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Jon Rambeau, president of Integrated Mission Systems at L3Harris, said the new vertical lift technology will enable long-range, crewed-uncrewed teaming for a range of missions. In this case, the gas-turbine hybrid VTOL aircraft could be used for low-altitude support missions, which could include contested logistics, electronic warfare, and counter unmanned aircraft systems, or C-UAS, Sciarra said.

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Kirsten Korosec

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Kirsten Korosec is a reporter and editor who has covered the future of transportation from EVs and autonomous...

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IMAGE CREDITS: JOBY AVIATION

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Joby Aviation is betting on hydrogen-electric aircraft for regional flight

Rebecca Bellan — 3:00 AM PDT · July 11, 2024

Joby Aviation is still a year away from commercially launching its electric air taxi designed for urban environments, but the startup is already looking toward its next chapter: intercity flight, powered by hydrogen.

To get the conversation started with regulators and demonstrate hydrogen's capabilities, Joby told TechCrunch that it completed a 523-mile test flight with a hydrogen-electric prototype aircraft — one of its eVTOLs (electric vertical takeoff and landing

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vehicle) that the startup fitted with a liquid hydrogen fuel cell and hydrogen-electric propulsion system.

Today, [Joby Aviation's](#) eVTOLs are being built to transport people and goods short distances within cities or from cities to airports. Powered by batteries, the aircraft have 100 miles of range. Hydrogen acts as a range-extender, opening the door to a regional use case, according to Joby.

"This is a landmark moment for aviation," Joe Ben Bevirt, CEO of Joby, told TechCrunch. "If you want to travel long distances, or you want to stay in the air for long periods of time, hydrogen-electric is a game changer."

The use of hydrogen to power vehicles has been hotly contested for years. Hydrogen is technically a zero emissions fuel source because it only emits water when in use. But it's expensive and energy intensive to produce, and most of today's stock is made using fossil fuels. Green hydrogen, which is produced via renewable energy sources, has yet to scale significantly.

However, recent private and public investment into green hydrogen, including an [\\$8 billion hydrogen hub program](#) within Biden's Inflation Reduction Act, is giving the sector new life. And Bevirt says "aviation has the potential to be a massive consumer of green hydrogen."

"The amazing thing about hydrogen is it's three times lighter than jet fuel per unit of energy. It's 100 times lighter than today's batteries per unit of energy," Bevirt said. "And with hydrogen fuel cells... we're able to convert the chemical energy in hydrogen into propulsion twice as efficiently."

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Joby has been quietly setting the stage to implement hydrogen into its eVTOLs. In 2021, Joby acquired H2Fly, a German hydrogen aviation startup. Last year, H2Fly demonstrated its own piloted flight with a liquid hydrogen-powered aircraft, and now, Joby has used that same technology to power a hydrogen version of its eVTOL. Bevirt told TechCrunch that the fuel cell system that H2Fly designed and built was able to power the six electric motors on the Joby aircraft and recharge its batteries in flight.

Joby is still very much in the demonstration phase, but when the startup is ready to start testing regional flights, it'll be able to integrate hydrogen-powered eVTOLs into its current system with minimal financial outlay, according to Bevirt.

"We take this battery electric aircraft we've built, and we take 90% of the systems and components that are in it, and we augment it with a hydrogen-electric range extender," said Bevirt. "And now, with a very small incremental investment, we can use the same vertiports, the same pilots and

mechanics and the Elevate operating system, which does all the back-end logic.”

ElevateOS, which is core to Joby’s planned air taxi service, is a nod to Uber’s Elevate air taxi business, which the ride-hail giant [sold to Joby in 2020](#). That sale included a set of software tools that enable on-demand mobility not unlike hailing an Uber ride. And in fact, [Joby’s app will be integrated with Uber](#) and Delta Airlines, the startup’s launch partner.

“So whether you pick up the Joby app or the Uber app, all of a sudden, it gives you the opportunity to not just go across town, but you can go anywhere in a region,” said Bevirt.

Some investors aren’t so convinced.

Cyrus Sigari, co-founder and managing partner of VC Up.Partners, said that while he’d love to see a hydrogen-powered eVTOL come to life, investors “would need to see a very compelling technical and business case to pursue investment in the category.”

One of the biggest challenges, he said, lies with the infrastructure.

“The industry is already scratching its head figuring out how to support battery electric aircraft with charging infrastructure at airports,” Sigari told TechCrunch. “Adding hydrogen filling stations into that equation will present even more challenges.”

He noted the recent [shutdown of Universal Hydrogen](#), one of the more prominent players in the hydrogen space for powering traditional airlines, has brought to light just how hard all of this is.

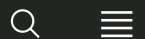
When asked about the hydrogen filling stations, Bevirt wasn’t fazed.

“We don’t expect that to be a significant impediment to the rollout,” Bevirt told TechCrunch. “We are in conversations with lots of airports across the country and around the world that are putting in liquid hydrogen refueling infrastructure.”

And even as Universal Hydrogen had to shutter because it was unable to raise enough funds to continue, [ZeroAvia signed a large deal with American Airlines](#), which committed to the purchase of 100 hydrogen-powered engines.

Joby didn’t provide a timeline for when it plans to launch its hydrogen-powered VTOL, but it did

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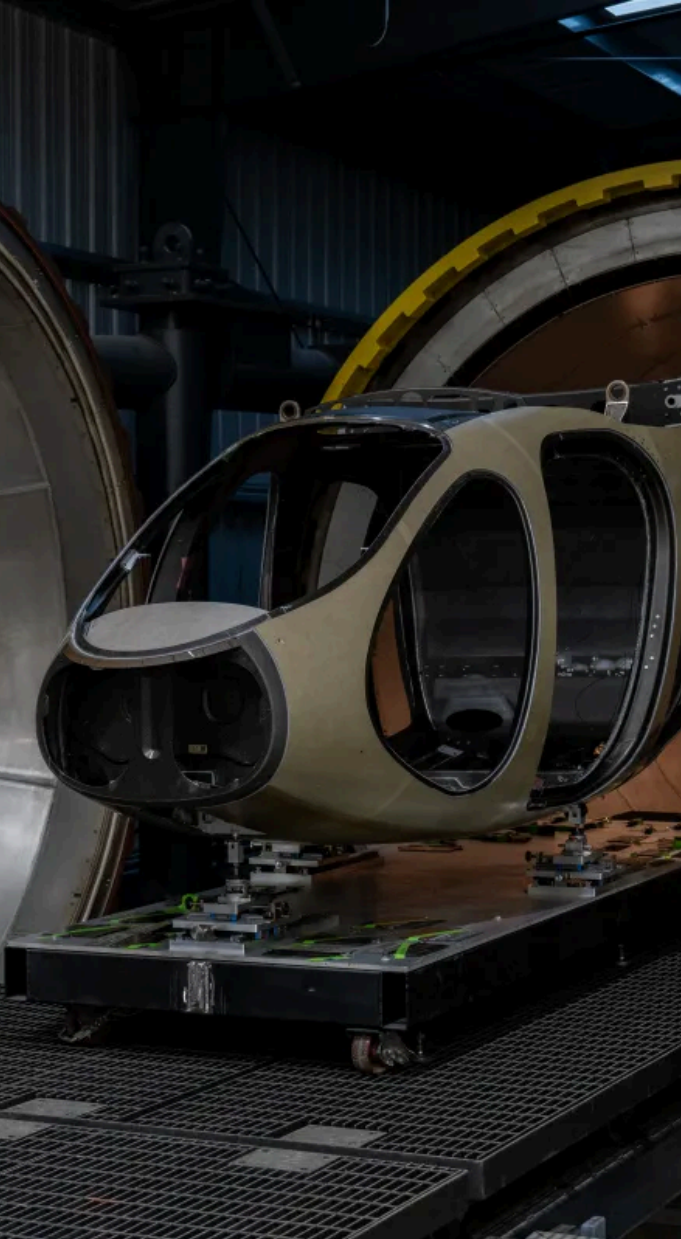


IMAGE CREDITS: JOBY AVIATION

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Joby flies towards 2025 commercial launch with final assembly of eVTOL

Rebecca Bellan — 5:00 AM PST · February 14, 2023

Electric vertical takeoff and landing (eVTOL) company Joby Aviation has taken another step on its path to start a commercial air taxi service by 2025. The company said Tuesday that it's begun final assembly on its "company-conforming" eVTOL, which is essentially a prototype that's a couple steps away from the final version.

Joby expects to begin flight testing for the aircraft in the first half of this year.

The technological milestone comes a week after [Joby completed the second stage](#) of a process with the Federal Aviation Administration (FAA) to achieve

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Kirsten Korosec - 12 hours ago

type certification for its aircraft for commercial passenger use. Joby says it's making moves on the final three stages of that process, as well.

Type certification would mean the FAA approves Joby's aircraft design and all component parts. It also signifies the design is compliant with the agency's standards for airworthiness and noise. After that, Joby will need to achieve production certification, which will allow Joby to mass produce its eVTOLs under the FAA-approved designs.

In getting to the point of final assembly, Joby had to put in place an intense quality management system that includes tracking and documenting every part on the aircraft, design drawings used to manufacture aircraft components, environmental conditions during fabrications and actions taken by manufacturing technicians.

Didier Papadopoulos, head of aircraft OEM at Joby, said the steps Joby has taken to produce a company-conforming aircraft, including the quality management system and regular FAA reviews, brings the company closer to receiving its production certification.

Joby has already built the major structures of the aircraft, like the wing, tail and fuselage, and will now fuse everything together and install wiring, electronics, actuation and propulsion systems.

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The aircraft will be the first one produced at Joby's pilot manufacturing facility in Marina, California. That facility is designed to prove out Joby's manufacturing process through low-rate manufacturing and is capable of producing tens of aircraft per year, according to the company. Joby is already using the facility to build and assemble parts for future aircraft.

When it comes to more scaled manufacturing, Joby is currently in the market for a location for its Phase 1 production facility. The company says it's actively reviewing proposals from a number of states keen to boost manufacturing and create jobs.

Joby Aviation secures certificate to operate commercial air taxis



Joby Aviation — the publicly traded electric air vehicle company with a current market cap of \$3.1 billion — has received the necessary certification from the Federal Aviation Administration to begin on-demand commercial air taxi operations, the company said on Thursday. While it's a significant milestone that puts Joby closer to its stated goal of ... [Continue reading](#)



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IMAGE CREDITS: JOBY AVIATION

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Joby Aviation receives permit to fly first eVTOL built on production line

Rebecca Bellan — 5:00 AM PDT · June 28, 2023

Joby Aviation has reached a series of milestones that put it on track to deliver its first electric aircraft to a customer by 2024 and commercialize an air taxi service by 2025.

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Joby's Marina, California pilot production plant, which operates in partnership with Toyota, has rolled out its first eVTOL (electric vertical takeoff and landing vehicle) — made possible by a helpful [\\$180 million injection](#) from investor Baillie Gifford in May. The aircraft is Joby's first production prototype, and it means Joby is one step away from having a commercially viable eVTOL for passenger use.

The production prototype eVTOL — Joby's third full-scale prototype — also received a special airworthiness certificate issued by the Federal Aviation Administration (FAA) and can now begin flight testing.

“This first aircraft coming off our pilot manufacturing line is a really, really big deal for the company,” Joe Ben Bevirt, founder and CEO of Joby, told TechCrunch. “We have spent many years building the processes and procedures and capabilities as an organization to be able to deliver this level of aircraft. We’ve been building since 2017, but to be able to build with this level of rigor is a huge step forward. It prepares us for this next stage of the certification process.”

Joby has already flown more than 30,000 miles since 2019 on pre-production prototype aircraft. This latest iteration builds on that experience and puts the company closer to achieving type certification with the FAA. Type certification means that the FAA approves of Joby's aircraft design and component parts, and that the design is compliant with the agency's standards for airworthiness and noise.

Joby nearly has three out of the five necessary stages of the type certification process underway, according to Bevirt. The final step after type certification would be production certification,

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which will allow Joby to mass produce eVTOLs under FAA-approved designs.

The eVTOL company already received its [Part 135 air carrier certificate](#), which allows Joby to begin on-demand commercial air taxi operations, in May 2022.



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Joby's production prototype aircraft has the potential to be the first eVTOL delivered to a customer, says the company. It will be deployed at Edwards Air Force Base in 2024, per [Joby's Agility Prime contract](#) with the U.S. Air Force, which is worth up to \$131 million. To meet that deadline, Joby will have to work quickly with the Air Force in order to secure military airworthiness certification.

The design of that aircraft will be replicated at Joby's pilot manufacturing facility.

"We have multiple of these aircraft coming down the production line, and that will be what we'll be

using both for customer applications and also for the type certification process,” said Bevirt.

Joby plans to commercialize an aerial rideshare service by 2025.

A deepening relationship with Toyota

Joby’s pilot manufacturing line was built in partnership with Toyota. The Japanese automaker is Joby’s largest external investor and will supply powertrains and actuation components to the company.

“Not only has Toyota invested almost \$400 million in the company, they’ve also put a significant number of engineers on site with us working shoulder to shoulder here in California, as well as having a team of engineers back in Japan working on processes,” said Bevirt. “So when people come through our pilot manufacturing line, they can see the incredible technologies and systems that Toyota has brought to bear and help Joby learn about how to do manufacturing.”

The pilot manufacturing line gives Joby the capability to make tens of aircraft per year, said Bevirt. The next stage is building a “phase one manufacturing facility” that will allow Joby to build larger quantities of eVTOL. Bevirt didn’t say where that manufacturing facility would be or when Joby hopes to break ground, but he did say the partnership with Toyota would be essential for that project, too.

In a move that deepens the relationship between the two companies, Joby also announced Wednesday that Tetsuo “Ted” Ogawa, president and CEO of Toyota Motor North America, will join Joby’s board of directors on July 1.

Ogawa has 39 years of experience at Toyota and currently leads Toyota North America as president and CEO, which runs 14 production plants in North America.

“As we look at scaling our manufacturing, we couldn’t be more thrilled to have somebody of his stature joining the Joby board,” said Bevirt. “We think this really demonstrates Toyota’s commitment to leading in this next stage of mobility.”

Aside from a manufacturer-investor relationship, Joby has worked with [Toyota and Japanese airline ANA](#) to explore ways to launch an air taxi service in Japan that connects to ground-based transportation.

Bervirt wouldn’t say where Joby intends to commercialize air taxi operations first, but he did say that the company has been working closely with the Japan Civil Aviation Bureau, which Bevirt describes as “extremely forward leaning.”

“We’ll be continuing to engage with them on our path to bringing the service to Japan,” he said.

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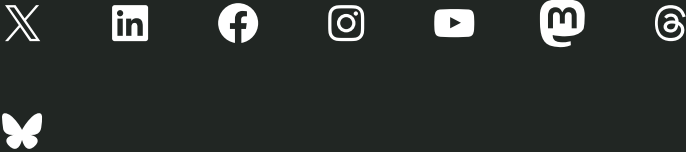
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